Collapse All

Online Lecture Room + Office Hour Links

[Lecture and Instructor Office Hours (https://canvas.wisc.edu/courses/478888/external_tools/27637)](https://canvas.wisc.edu/courses/478888/external_tools/27637)

[TA Office Hours (https://canvas.wisc.edu/courses/478888/external_tools/27637)](https://canvas.wisc.edu/courses/478888/external_tools/27637)

Lecture 1 - 09/04 - Introduction and Optical Illusions

Quiz - Introduction
Sep 6 0.5 pts

HW1
Sep 16 5 pts

Slide

[Lecture 1: Introduction and Optical Illusions (https://drive.google.com/file/d/1YegzPtnTO1tYGmy0cHGGi7lCHoNYA6Qq/view?usp=sharing)](https://drive.google.com/file/d/1YegzPtnTO1tYGmy0cHGGi7lCHoNYA6Qq/view?usp=sharing)

Lecture 2 - 09/09 - Image Formation and Sensing

Quiz - Image Formation and Sensing
Sep 11 0.5 pts

Videos

[Video 1: Intro and Perspective Imaging (https://drive.google.com/file/d/1py9JP2ckzJsSO9aZmQH9AoauEPsmFFdX/view?usp=sharing)]

[Video 2: Magnification and Vanishing Point (https://drive.google.com/file/d/1n2eW733G-YhODvgRDXiv-sJaioqxRRLJ/view?usp=sharing)]

[Video 3: Lenses (https://drive.google.com/file/d/1-R_qDzgspwc5_YCgjsi0V4NzeJt35wFO/view?usp=sharing)]

[Video 4: Defocus (https://drive.google.com/file/d/1gKStcg48QBy4OaEYU9oq6MyTEAdvJiEb/view?usp=sharing)]

[Video 5: Lens-based Image Artifacts (https://drive.google.com/file/d/1aJnEWKtnKHyTBjeR-UAbg76kTegQjHWD/view?usp=sharing)]

[Video 6: Image Sensors, Noise, and Dynamic Range (https://drive.google.com/file/d/1z4V4iFkacDUbOYx8i0gouQLcW1oN3XgM/view?usp=sharing)]

Slide

[Lecture 2: Image Formation and Sensing (https://drive.google.com/file/d/19ROy1qZgFq9nQPu3PMMmTGIXiT7iOEPm/view?usp=sharing)]

Lecture 3 - 09/11 - Image Filtering

Quiz - Image Filtering

Sep 13 0.5 pts

Videos

[Video 1: Intro (https://drive.google.com/file/d/1IQVt0OEUUqeVjpfgRNeyjG_6j64UW52N/view?usp=sharing)](https://drive.google.com/file/d/1IQVt0OEUUqeVjpfgRNeyjG_6j64UW52N/view?usp=sharing)

[Video 2: Linear Filtering (https://drive.google.com/file/d/1kD7dDewYppHl3y4Q9sGjGSrtfApzgJjl/view?usp=sharing)](https://drive.google.com/file/d/1kD7dDewYppHl3y4Q9sGjGSrtfApzgJjl/view?usp=sharing)

[Video 3: Box Filter Example Linear Filtering (https://drive.google.com/file/d/1dH3nDw_a6DW6S66PoGRSIK22Of7uo2t8/view?usp=sharing)](https://drive.google.com/file/d/1dH3nDw_a6DW6S66PoGRSIK22Of7uo2t8/view?usp=sharing)

[Video 4: Filtering Examples and Gaussian Smoothing (https://drive.google.com/file/d/1Fu8mF9MQ2amubzK60FO_G_hU0NEiJE71/view?usp=sharing)](https://drive.google.com/file/d/1Fu8mF9MQ2amubzK60FO_G_hU0NEiJE71/view?usp=sharing)

[Video 5: Median Filtering (https://drive.google.com/file/d/1HEPGwRlk2HYCw11ybZmXdRMBLGgTXv8A/view?usp=sharing)](https://drive.google.com/file/d/1HEPGwRlk2HYCw11ybZmXdRMBLGgTXv8A/view?usp=sharing)

[Video 6: Template Matching (https://drive.google.com/file/d/1nWczVoshrrNiplg-rRNXslL5kL8DAtdD/view?usp=sharing)](https://drive.google.com/file/d/1nWczVoshrrNiplg-rRNXslL5kL8DAtdD/view?usp=sharing)

Slide

[Lecture 3: Image Filtering (https://drive.google.com/file/d/10Sstb1q2z80PvUyH_iZhYXil_vPLh8he/view?usp=sharing)](https://drive.google.com/file/d/10Sstb1q2z80PvUyH_iZhYXil_vPLh8he/view?usp=sharing)

▾ Lecture 4 - 09/16 - Thinking about Images in the Fourier Domain

Quiz -Thinking about Images in the Fourier Domain
Sep 18 0.5 pts

Videos

[Video 1: Intro to Fourier Transform (https://drive.google.com/file/d/1YrWmW9SRYS4O7d8emnKfhMICFPIU-nTf/view?usp=sharing)](https://drive.google.com/file/d/1YrWmW9SRYS4O7d8emnKfhMICFPIU-nTf/view?usp=sharing)

[Video 2: Fourier Transform Examples (https://drive.google.com/file/d/1F9UqAGmrmxZ-3VU4xqL9plqDm4hSu7DH/view?usp=sharing)](https://drive.google.com/file/d/1F9UqAGmrmxZ-3VU4xqL9plqDm4hSu7DH/view?usp=sharing)

[Video 3: Fourier Transform Properties (https://drive.google.com/file/d/1OlnZqdfp7SuvrvduPqlrH4EP2BDFhAXF/view?usp=sharing)](https://drive.google.com/file/d/1OlnZqdfp7SuvrvduPqlrH4EP2BDFhAXF/view?usp=sharing)

[Video 4: 2D Fourier Transform Example (https://drive.google.com/file/d/1hiN-kacix-okd3aCWOxpgWHQj55Seqe6/view?usp=sharing)](https://drive.google.com/file/d/1hiN-kacix-okd3aCWOxpgWHQj55Seqe6/view?usp=sharing)

[Video 5: Filtering in the Fourier Domain (https://drive.google.com/file/d/1PVXvU38eA2yVdDj8g3pN74h72B0Err8d/view?usp=sharing)](https://drive.google.com/file/d/1PVXvU38eA2yVdDj8g3pN74h72B0Err8d/view?usp=sharing)

[Video 6: Deconvolution (https://drive.google.com/file/d/1Wl_AXvAdTxM-IS5ko2AB9tZPD9tcmK-V/view?usp=sharing)](https://drive.google.com/file/d/1Wl_AXvAdTxM-IS5ko2AB9tZPD9tcmK-V/view?usp=sharing)

[Video 7: Sampling (https://drive.google.com/file/d/1OF89U-aXdAku6e31clag_4nYWbiR_8yg/view?usp=sharing)](https://drive.google.com/file/d/1OF89U-aXdAku6e31clag_4nYWbiR_8yg/view?usp=sharing)

Slide

[Lecture 4: Thinking about Images in the Fourier Domain (https://drive.google.com/file/d/1XA96olLFhR5XZ5RuniYB8li68KEfFV25/view?usp=sharing)](https://drive.google.com/file/d/1XA96olLFhR5XZ5RuniYB8li68KEfFV25/view?usp=sharing)

Lecture 5 - 09/18 - Edge and Corner Detection

Quiz - Edge and Corner Detection

Sep 20 0.5 pts

Videos

[Video 1: Intro (https://drive.google.com/file/d/1JdWQJ5oYv-7Thf8mOEUK2qK8V6CuS4qm/view?usp=sharing)](https://drive.google.com/file/d/1JdWQJ5oYv-7Thf8mOEUK2qK8V6CuS4qm/view?usp=sharing)

[Video 2: Theory of Gradient Edge Detector (https://drive.google.com/file/d/1zU48w4P2K2AZAm1uSS7Rg-6J4vZz3coP/view?usp=sharing)](https://drive.google.com/file/d/1zU48w4P2K2AZAm1uSS7Rg-6J4vZz3coP/view?usp=sharing)

[Video 3: Discrete Gradient Operator (https://drive.google.com/file/d/1qp_y3xr833CW5Gbn9XDDnkPFK0EJxd18/view?usp=sharing)](https://drive.google.com/file/d/1qp_y3xr833CW5Gbn9XDDnkPFK0EJxd18/view?usp=sharing)

[Video 4: Laplacian and Noise Suppression (https://drive.google.com/file/d/1leePnVUHpYIqLSFR1g1FtENOzwqpqMZ7/view?usp=sharing)](https://drive.google.com/file/d/1leePnVUHpYIqLSFR1g1FtENOzwqpqMZ7/view?usp=sharing)

[Video 5: Canny Edge Detector (https://drive.google.com/file/d/1C39Wh8voTLCFEh9rTZJSpsQCOZB7jK44/view?usp=sharing)](https://drive.google.com/file/d/1C39Wh8voTLCFEh9rTZJSpsQCOZB7jK44/view?usp=sharing)

[Video 6: Harris Corner Detection (https://drive.google.com/file/d/10PksSdFL-tdhU7iGZsscmK_D8j-3Jiy0/view?usp=sharing)](https://drive.google.com/file/d/10PksSdFL-tdhU7iGZsscmK_D8j-3Jiy0/view?usp=sharing)

Slide

[Lecture 5: Edge and Corner Detection (https://drive.google.com/file/d/1JBBDRxbwbQeO1tNzGXgnVOX7wjWPMLC-/view?usp=sharing)](https://drive.google.com/file/d/1JBBDRxbwbQeO1tNzGXgnVOX7wjWPMLC-/view?usp=sharing)

Lecture 6 & Lecture 7 - 09/23 & 9/25 - Boundary Detection and Hough Transforms & Video …

HW2
Oct 7 16 pts

Quiz - Image Formation, Processing, and Edge Detection Review
Sep 25 0.5 pts

Quiz - Boundary Detection and Hough Transforms
Sep 27 0.5 pts

Lecture Videos

[Video 1: Intro (https://drive.google.com/file/d/1rDbK2V5_zdRYsjUYgaUT33G1EATf7qQv/view?usp=sharing)](https://drive.google.com/file/d/1rDbK2V5_zdRYsjUYgaUT33G1EATf7qQv/view?usp=sharing)

[Video 2: Line Fitting (https://drive.google.com/file/d/1-BdIUcnT-9sP90LUY3MYVTEajVQ2RYyG/view?usp=sharing)](https://drive.google.com/file/d/1-BdIUcnT-9sP90LUY3MYVTEajVQ2RYyG/view?usp=sharing)

[Video 3: Hough Line (https://drive.google.com/file/d/1z17-BI71H2q3gPcrXPkspVsTUoMbEkAB/view?usp=sharing)](https://drive.google.com/file/d/1z17-BI71H2q3gPcrXPkspVsTUoMbEkAB/view?usp=sharing)

[Video 4: Hough Transform Practical Considerations (https://drive.google.com/file/d/1EHIJHb4iEHZm1cMG--m-5KibXZqwqWvs/view?usp=sharing)](https://drive.google.com/file/d/1EHIJHb4iEHZm1cMG--m-5KibXZqwqWvs/view?usp=sharing)

[Video 5: Circle and Generalized Hough Transform (https://drive.google.com/file/d/1fqSLV4tnGaH6N749iNr1YlkdY2KT8Gtd/view?usp=sharing)](https://drive.google.com/file/d/1fqSLV4tnGaH6N749iNr1YlkdY2KT8Gtd/view?usp=sharing)

[Video 6: Active Contours Intro (https://drive.google.com/file/d/1zDjyzf22VocZCFHI_1a20LFI9wHe_Q7-/view?usp=sharing)](https://drive.google.com/file/d/1zDjyzf22VocZCFHI_1a20LFI9wHe_Q7-/view?usp=sharing)

[Video 7: Active Contours Optimization (https://drive.google.com/file/d/1rrnLa4xbP7DoXub0T8kjN2Fr4p4ernv5/view?usp=sharing)](https://drive.google.com/file/d/1rrnLa4xbP7DoXub0T8kjN2Fr4p4ernv5/view?usp=sharing)

Slide

[Lecture 6 & 7 Boundary Detection and Hough Transforms & Video Boundary Detection using Active Contours (https://drive.google.com/file/d/1dj8SWS3zoNS2EhJ1ljVcinHLZVxo066O/view?usp=sharing)](https://drive.google.com/file/d/1dj8SWS3zoNS2EhJ1ljVcinHLZVxo066O/view?usp=sharing)

Lecture 8 - 09/30 - 2D Object Recognition using SIFT

No Quiz for Lecture 8

Videos

[Video 1: Intro to Image Matching (https://drive.google.com/file/d/1EYEUIAcR8svX_FbcX6oAYYTZhCcmicB-/view?usp=sharing)](https://drive.google.com/file/d/1EYEUIAcR8svX_FbcX6oAYYTZhCcmicB-/view?usp=sharing)

[Video2: 1D Blob Detection (https://drive.google.com/file/d/1F8rm4xOnxf7KbQI9PKpZjfUt3Uut8OiZ/view?usp=sharing)](https://drive.google.com/file/d/1F8rm4xOnxf7KbQI9PKpZjfUt3Uut8OiZ/view?usp=sharing)

[Video 3: 2D Blob Detection (https://drive.google.com/file/d/1TvDNYffld0GwZp49WKVEt7Tyq4ZjTdCJ/view?usp=sharing)](https://drive.google.com/file/d/1TvDNYffld0GwZp49WKVEt7Tyq4ZjTdCJ/view?usp=sharing)

[Video 4: Scale Space Analysis (https://drive.google.com/file/d/11hsay12JVJ0R6X50r9oYxJE8o8WiLpDi/view?usp=sharing)](https://drive.google.com/file/d/11hsay12JVJ0R6X50r9oYxJE8o8WiLpDi/view?usp=sharing)

[Video 5: SIFT Descriptor (https://drive.google.com/file/d/17GZyKVjGW2LCeYd-XJLKNLgNmEK_-Bm_/view?usp=sharing)](https://drive.google.com/file/d/17GZyKVjGW2LCeYd-XJLKNLgNmEK_-Bm_/view?usp=sharing)

[Video 6: SIFT Examples and Active Contours (https://drive.google.com/file/d/1uiv2_iLSnTYM-gcOMJSKqly3-D1q-ARs/view?usp=sharing)](https://drive.google.com/file/d/1uiv2_iLSnTYM-gcOMJSKqly3-D1q-ARs/view?usp=sharing)

Slide

[Lecture 8 2D Object Recognition using SIFT (https://drive.google.com/file/d/1oUQLi-cMmjQIS7vtfhLADhyorF-tR1hn/view?usp=sharing)](https://drive.google.com/file/d/1oUQLi-cMmjQIS7vtfhLADhyorF-tR1hn/view?usp=sharing)

▾ Lecture 09 and 10 - 10/02 and 10/07 - Image Alignment: Image Transformations and RANSAC

HW3
Oct 23 13 pts

Survey about Quizzes
Dec 8

Quiz - Image Alignment
Oct 4 0.5 pts

Videos

[Video 1: Intro and Overview (https://drive.google.com/file/d/1jplfVRR6ZFGPZpZcXAGPhIdHQECDYzVm/view?usp=sharing)](https://drive.google.com/file/d/1jplfVRR6ZFGPZpZcXAGPhIdHQECDYzVm/view?usp=sharing)

[Video 2: Linear Warping (https://drive.google.com/file/d/1SMOteesTEw-NY_tGoeMeQazHp0AuJvXF/view?usp=sharing)](https://drive.google.com/file/d/1SMOteesTEw-NY_tGoeMeQazHp0AuJvXF/view?usp=sharing)

[Video 3: Homogeneous Coordinates (https://drive.google.com/file/d/1I0PSecYHr9fN1gLh5-wdxxMOvYeozKeS/view?usp=sharing)](https://drive.google.com/file/d/1I0PSecYHr9fN1gLh5-wdxxMOvYeozKeS/view?usp=sharing)

[Video 4: Homography (https://drive.google.com/file/d/18VWBnM0F13zkUz3qQRT5nHEmr0CBvr8V/view?usp=sharing)](https://drive.google.com/file/d/18VWBnM0F13zkUz3qQRT5nHEmr0CBvr8V/view?usp=sharing)

[Video 5: Least Squares Homography (https://drive.google.com/file/d/1vFXPOR646b9TxEmrxV_Wo9fSWtq0PPgC/view?usp=sharing)](https://drive.google.com/file/d/1vFXPOR646b9TxEmrxV_Wo9fSWtq0PPgC/view?usp=sharing)

[Video 6: RANSAC (https://drive.google.com/file/d/1z48_p8INvSYZ3QJ45NdOuGsBowY1e_-u/view?usp=sharing)](https://drive.google.com/file/d/1z48_p8INvSYZ3QJ45NdOuGsBowY1e_-u/view?usp=sharing)

[Video 7: Warping (https://drive.google.com/file/d/1ayn_v57W7FxxPF1TKvZ_g47LfyqXFt9y/view?usp=sharing)](https://drive.google.com/file/d/1ayn_v57W7FxxPF1TKvZ_g47LfyqXFt9y/view?usp=sharing)

[Video 8: Blending (https://drive.google.com/file/d/1aFC-mmhtphMpRb7uMf5NGY1HRb-y6BV-/view?usp=sharing)](https://drive.google.com/file/d/1aFC-mmhtphMpRb7uMf5NGY1HRb-y6BV-/view?usp=sharing)

Slide

[Lecture 9 & 10 Image Alignment: Image Transformations and RANSAC (https://drive.google.com/file/d/1zCpTBA26FtLFKfuvHNVmoG3oLerULMFT/view?usp=sharing)](https://drive.google.com/file/d/1zCpTBA26FtLFKfuvHNVmoG3oLerULMFT/view?usp=sharing)

Lecture 11 - 10/09 - Face Detection with Support Vector Machines

Quiz - Face Detection and SVM
Oct 14 0.5 pts

Videos

Video 1: Intro to Face Detection

Video 2: Haar Features

Video 3: Integral Images

Video 4: Nearest Neighbor Classifier

Video 5: Linear Decision Boundaries

Video 6: Support Vector Machines

Slide

[Lecture 11 Face Detection with Support Vector Machines (https://drive.google.com/file/d/1yLJH0080tZCTt3TC2UNk8t_N3aFm8PNr/view?usp=sharing)](https://drive.google.com/file/d/1yLJH0080tZCTt3TC2UNk8t_N3aFm8PNr/view?usp=sharing)

Lecture 12 - 10/14 - Image Segmentation with k-Means Clustering and Mean-Shift

No Quiz for this Lecture

Videos

[Video 1: Intro (https://drive.google.com/file/d/1uk0Nb3jiNM0LeWzUMb10XHx5x7egyGig/view?usp=sharing)](https://drive.google.com/file/d/1uk0Nb3jiNM0LeWzUMb10XHx5x7egyGig/view?usp=sharing)

[Video 2: Clustering (https://drive.google.com/file/d/1FLS8b9t5oWU02m3memHO46wRBzGLNrTb/view?usp=sharing)](https://drive.google.com/file/d/1FLS8b9t5oWU02m3memHO46wRBzGLNrTb/view?usp=sharing)

[Video 3: K-Means Clustering (https://drive.google.com/file/d/19ShI7vkYIlOMJe_IFbz1_Vx-XqzG_9Eo/view?usp=sharing)](https://drive.google.com/file/d/19ShI7vkYIlOMJe_IFbz1_Vx-XqzG_9Eo/view?usp=sharing)

[Video 4: K Means Segmentation (https://drive.google.com/file/d/1nPP1C-UbDBtjM42J84qnsFZhxiB0NsEM/view?usp=sharing)](https://drive.google.com/file/d/1nPP1C-UbDBtjM42J84qnsFZhxiB0NsEM/view?usp=sharing)

[Video 5: Mean Shift Clustering (https://drive.google.com/file/d/1iq1olNrryzumENOy0IM8hENvCkvzPrqg/view?usp=sharing)](https://drive.google.com/file/d/1iq1olNrryzumENOy0IM8hENvCkvzPrqg/view?usp=sharing)

[Video 6: Mean Shift Segmentation (https://drive.google.com/file/d/14oVnc-cOC1wAM2NNHPJTkrlmoKduwdzl/view?usp=sharing)](https://drive.google.com/file/d/14oVnc-cOC1wAM2NNHPJTkrlmoKduwdzl/view?usp=sharing)

[Video 7: Graph Cut Segmentation (https://drive.google.com/file/d/1KDm5HlWzK4xD2TtkaQ0RISD45WxtdUog/view?usp=sharing)](https://drive.google.com/file/d/1KDm5HlWzK4xD2TtkaQ0RISD45WxtdUog/view?usp=sharing)

[Video 8: Normalized Cut Segmentation (https://drive.google.com/file/d/1Z70gxD7u1BmHXdOs-SeejAD3ajkelnV9/view?usp=sharing)](https://drive.google.com/file/d/1Z70gxD7u1BmHXdOs-SeejAD3ajkelnV9/view?usp=sharing)

Slide

[Lecture 12 Image Segmentation with k-Means Clustering and Mean-Shift (https://drive.google.com/file/d/1yUMqEtEL9ZfB-e83pvFylrpBBQgzcUd-/view?usp=sharing)](https://drive.google.com/file/d/1yUMqEtEL9ZfB-e83pvFylrpBBQgzcUd-/view?usp=sharing)

Lecture 13 - 10/16 - Introduction to Neural Networks and Convolutional Neural Networks

Quiz - Introduction to Neural Networks
Oct 22 0.5 pts

Videos

[01 - Introduction (https://drive.google.com/file/d/1zDztKoNlMhFXrKSNTe5cRhLmJc4eVVHE/view?usp=sharing)](https://drive.google.com/file/d/1zDztKoNlMhFXrKSNTe5cRhLmJc4eVVHE/view?usp=sharing)

[02 - Neurons and Activations (https://drive.google.com/file/d/1ton280boWNkVfCRb7EWDjWV7NtuusW9a/view?usp=sharing)](https://drive.google.com/file/d/1ton280boWNkVfCRb7EWDjWV7NtuusW9a/view?usp=sharing)

[03 - Character Recognition Example (https://drive.google.com/file/d/1dh2nd6GznI50qDrXkhlyGhaEon5DJ0Ck/view?usp=sharing)](https://drive.google.com/file/d/1dh2nd6GznI50qDrXkhlyGhaEon5DJ0Ck/view?usp=sharing)

[04 - Neural Network Training (https://drive.google.com/file/d/1gwaoTtF97Wsd8T43r2JmyL8xPgyVtjPF/view?usp=sharing)](https://drive.google.com/file/d/1gwaoTtF97Wsd8T43r2JmyL8xPgyVtjPF/view?usp=sharing)

[05 - Motivating Convolutional Networks (https://drive.google.com/file/d/1-lvWVWYCR4PFmMlvkwg_X3bdu4aA7Fc4/view?usp=sharing)](https://drive.google.com/file/d/1-lvWVWYCR4PFmMlvkwg_X3bdu4aA7Fc4/view?usp=sharing)

[06 - Convolutional Networks Description (https://drive.google.com/file/d/1TXcKYhPTMHJwQS0a0KUlXiyFOsCsGCUK/view?usp=sharing)](https://drive.google.com/file/d/1TXcKYhPTMHJwQS0a0KUlXiyFOsCsGCUK/view?usp=sharing)

[07 - ConvNets for Semantic Segmentation (https://drive.google.com/file/d/1fKZX_LVX5kvE4CpCm6PlkU1NqsF3DlkZ/view?usp=sharing)](https://drive.google.com/file/d/1fKZX_LVX5kvE4CpCm6PlkU1NqsF3DlkZ/view?usp=sharing)

Slide

[Lecture 13 Intro to Neural Networks and Convolutional Neural Networks (https://drive.google.com/file/d/1H3wXPHw0_Dwo6K36cllm9LsJ4Sud6U9a/view?usp=sharing)](https://drive.google.com/file/d/1H3wXPHw0_Dwo6K36cllm9LsJ4Sud6U9a/view?usp=sharing)

▾ Lecture 14 & 15- 10/23 & 10/28 - Radiometry and Surface Appearance & 3D Shape from Pho…

Quiz - Radiometry and Photometric Stereo
Oct 30 0.5 pts

Videos

[01 - Introduction to Radiometry (https://drive.google.com/file/d/1ZlO38-K9eL_zq-tXitYwz_juJ7bTQF4D/view?usp=sharing)](https://drive.google.com/file/d/1ZlO38-K9eL_zq-tXitYwz_juJ7bTQF4D/view?usp=sharing)

[02 - BRDF (https://drive.google.com/file/d/1BwVWV5Ymx_akzYsx0-AjH6FYqrDEA5nA/view?usp=sharing)](https://drive.google.com/file/d/1BwVWV5Ymx_akzYsx0-AjH6FYqrDEA5nA/view?usp=sharing)

[03 - Lambertian Model (https://drive.google.com/file/d/1LqGLbqRxtD8mSDkGJURk1D9mfGqVDw0M/view?usp=sharing)](https://drive.google.com/file/d/1LqGLbqRxtD8mSDkGJURk1D9mfGqVDw0M/view?usp=sharing)

[04 - Specular and Rough Surfaces (https://drive.google.com/file/d/1CGA2htWEaaNGqwPHG2xyc4KshbVneggp/view?usp=sharing)](https://drive.google.com/file/d/1CGA2htWEaaNGqwPHG2xyc4KshbVneggp/view?usp=sharing)

[05 - Photometric Stereo (https://drive.google.com/file/d/1z-Qz9bJaswW_q4ixNi6xs7qNjwZnhQbl/view?usp=sharing)](https://drive.google.com/file/d/1z-Qz9bJaswW_q4ixNi6xs7qNjwZnhQbl/view?usp=sharing)

[06 - Photometric Stereo Analysis and Applications (https://drive.google.com/file/d/19_76YApJOc5HHwPPrZ3nmS6KzC__rCZ1/view?usp=sharing)](https://drive.google.com/file/d/19_76YApJOc5HHwPPrZ3nmS6KzC__rCZ1/view?usp=sharing)

[07 - Example-based Photometric Stereo (https://drive.google.com/file/d/1HZFJBAFRdI7lxdCzMypEzXtlflo652Qf/view?usp=sharing)](https://drive.google.com/file/d/1HZFJBAFRdI7lxdCzMypEzXtlflo652Qf/view?usp=sharing)

Slide

[Lecture 14 15 Radiometry and Surface Appearance 3D Shape from Photometric Stereo (https://drive.google.com/file/d/1jRCzU5bHWHfQmqij8XiTK3WJfprL0Dsg/view?usp=sharing)](https://drive.google.com/file/d/1jRCzU5bHWHfQmqij8XiTK3WJfprL0Dsg/view?usp=sharing)

Lecture 16 - 10/30 - Shape from Focus/Defocus

Quiz - Shape from Focus/Defocus
Nov 4 0.5 pts

HW4
Nov 13 7 pts

Videos

[01 - Introduction to Camera Defocus (https://drive.google.com/file/d/1qfa9HD2jtTkv6vYKXiBBqqee4jUkQ8sl/view?usp=sharing)](https://drive.google.com/file/d/1qfa9HD2jtTkv6vYKXiBBqqee4jUkQ8sl/view?usp=sharing)

[02 - Depth from Focus (https://drive.google.com/file/d/1rB1RsLKcE82Dh2zGDy-Cm7iE-MxXf7gu/view?usp=sharing)](https://drive.google.com/file/d/1rB1RsLKcE82Dh2zGDy-Cm7iE-MxXf7gu/view?usp=sharing)

[03 - Depth from Defocus (https://drive.google.com/file/d/1PrC-1EMr18Wwl99hKnUbgSKPGKGnK4le/view?usp=sharing)](https://drive.google.com/file/d/1PrC-1EMr18Wwl99hKnUbgSKPGKGnK4le/view?usp=sharing)

Slide

[Lecture 16 Shape from Focus/Defocus (https://drive.google.com/file/d/1Zb-rNSK5WWSTGTaA7D_GWNY04VgO7vOx/view?usp=sharing)](https://drive.google.com/file/d/1Zb-rNSK5WWSTGTaA7D_GWNY04VgO7vOx/view?usp=sharing)

Lecture 17 - 11/04 - Camera Calibration and Shape from Stereo

Quiz - Camera Calibration and Simple Stereo
Nov 11 0.5 pts

Videos

[01 - Introduction to Depth from Stereo (https://drive.google.com/file/d/1iXVVfBu3hxVWcVYclNd2iRT2h_hRS-2j/view?usp=sharing)](https://drive.google.com/file/d/1iXVVfBu3hxVWcVYclNd2iRT2h_hRS-2j/view?usp=sharing)

[02 - Intrinsic Camera Matrix (https://drive.google.com/file/d/1FDcg6OH1wHQx3rxzYmAVj9lCa0COoCcd/view?usp=sharing)](https://drive.google.com/file/d/1FDcg6OH1wHQx3rxzYmAVj9lCa0COoCcd/view?usp=sharing)

[03 - Extrinsic Camera Matrix (https://drive.google.com/file/d/1IJKqi_Gx1sAXAKV_RHCQy8VSE1KoMNPq/view?usp=sharing)](https://drive.google.com/file/d/1IJKqi_Gx1sAXAKV_RHCQy8VSE1KoMNPq/view?usp=sharing)

[04 - Camera Calibration (https://drive.google.com/file/d/1Al7eElLBPf-RWZ7dRoA1cWKYuqK8-Ff3/view?usp=sharing)](https://drive.google.com/file/d/1Al7eElLBPf-RWZ7dRoA1cWKYuqK8-Ff3/view?usp=sharing)

[05 - Recovering Extrinsic and Intrinsic Matrices (https://drive.google.com/file/d/1ltWqA8w4VMS5qVbjXTZ6_x2PrM0eWKwF/view?usp=sharing)](https://drive.google.com/file/d/1ltWqA8w4VMS5qVbjXTZ6_x2PrM0eWKwF/view?usp=sharing)

[06 - Triangulation (https://drive.google.com/file/d/1axNE8xQdAtXhJX-SUimgArHJI7nZ_Ooo/view?usp=sharing)](https://drive.google.com/file/d/1axNE8xQdAtXhJX-SUimgArHJI7nZ_Ooo/view?usp=sharing)

[07 - Finding Correspondences (https://drive.google.com/file/d/13Wpk1dbnlA3fbb9W_84Wa5Okl68vMesX/view?usp=sharing)](https://drive.google.com/file/d/13Wpk1dbnlA3fbb9W_84Wa5Okl68vMesX/view?usp=sharing)

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[Lecture 17 Camera Calibration and Shape from Stereo (https://drive.google.com/file/d/1Z_JQWDPbO7vomjBc8qV6xA2bVXO5YuHQ/view?usp=sharing)](https://drive.google.com/file/d/1Z_JQWDPbO7vomjBc8qV6xA2bVXO5YuHQ/view?usp=sharing)

Lecture 18 - 11/06 - Internet-Scale 3D Reconstruction

Quiz - Internet-Scale 3D Reconstruction

Nov 11 0.5 pts

Videos

[01 - Introduction and Review of Basic Stereo (https://drive.google.com/file/d/1FlfMa9R7yXqh9L_5MYbxx8F6DWqJlUiX/view?usp=sharing)](https://drive.google.com/file/d/1FlfMa9R7yXqh9L_5MYbxx8F6DWqJlUiX/view?usp=sharing)

[02 - Intuition and Outline (https://drive.google.com/file/d/1-HAbfFY2WDdSE8bA54n5dO_Y1l9K6l2x/view?usp=sharing)](https://drive.google.com/file/d/1-HAbfFY2WDdSE8bA54n5dO_Y1l9K6l2x/view?usp=sharing)

[03 - Essential Matrix (https://drive.google.com/file/d/15gEN8WP4nBiEOQTiaM5LHLTGlz8qt241/view?usp=sharing)](https://drive.google.com/file/d/15gEN8WP4nBiEOQTiaM5LHLTGlz8qt241/view?usp=sharing)

[04 - Fundamental Matrix (https://drive.google.com/file/d/1U68XSNGguKjv-rzvGjfeXWewDy6RWxMy/view?usp=sharing)](https://drive.google.com/file/d/1U68XSNGguKjv-rzvGjfeXWewDy6RWxMy/view?usp=sharing)

[05 - Stereo Autocalibration (https://drive.google.com/file/d/1R7eQvzZ-5hF9uWkrKljGu_9-_GEx7aFt/view?usp=sharing)](https://drive.google.com/file/d/1R7eQvzZ-5hF9uWkrKljGu_9-_GEx7aFt/view?usp=sharing)

[06 - Finding Dense Correspondences via Epipolar Geometry (https://drive.google.com/file/d/1vq3tVERU9EZsOamq4-EGjC8dM9D8lbJq/view?usp=sharing)](https://drive.google.com/file/d/1vq3tVERU9EZsOamq4-EGjC8dM9D8lbJq/view?usp=sharing)

[07 - Triangulation and Internet Scale 3D (https://drive.google.com/file/d/12eUmpL3jPd2uyhOec6P9-AotJ2VmGY4d/view?usp=sharing)](https://drive.google.com/file/d/12eUmpL3jPd2uyhOec6P9-AotJ2VmGY4d/view?usp=sharing)

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[Lecture 18 Internet-Scale 3D Reconstruction (https://drive.google.com/file/d/1Z_0GW7me7uAT6P0vVjfGnkejWm24F7nA/view?usp=sharing)](https://drive.google.com/file/d/1Z_0GW7me7uAT6P0vVjfGnkejWm24F7nA/view?usp=sharing)

Lecture 19 - 11/11 - Optical Flow and Motion Measurement

Quiz - Optical Flow and Motion
Nov 13 0.5 pts

Videos

01 - Introduction to Motion Estimation

02 - Theory of Optical Flow

03 - Lucas-Kanade Optical Flow

04 - Large-Motion Optical Flow

Will unlock Nov 10 at 5pm

Lecture 20 - 11/13 - Background Subtraction and Image Tracking

Quiz - Object Tracking
Nov 18 0.5 pts

Videos

01 - Introduction and Change Detection

02 - Motion Detection using Gaussian Mixture Models

03 - Template-based Tracking

04 - Detection-based Tracking

Will unlock Nov 12 at 5pm

Lecture 21 - 11/18 - Modern 3D Cameras: Structured Light and Time-of-Flight

Quiz - Structured Light and Time-of-Flight
Nov 20 0.5 pts

Will unlock Nov 17 at 5pm

Lecture 22 - 11/20 - Advanced Topics: Computational Cameras

Will unlock Nov 20 at 12am